



NUMBER SYSTEM

* When any two digit number having the digit sum is always divided by 11 & difference of number always divided by 9.

* Any 2 digit number repeated 2 times is always divided by smallest 2 digit prime no. (101).

* Sum of n natural no. $\therefore \frac{n(n+1)}{2}$

* Sum of random sequence of no. $\therefore \frac{n}{2}(a+l)$

* Sum of squares of first n natural no. $\therefore \frac{n(n+1)(2n+1)}{6}$ $[n = \text{last No.}]$

* Sum of squares of even/odd no. $\therefore \frac{n(n+1)(n+2)}{6}$

* Sum of cubes of odd no. $\therefore \frac{(n+1)^2 [(n+1)^2 - 2]}{8}$

* Sum of cubes of even no. $\therefore 2 \times \frac{n}{2} \left[\frac{(n+1)^2}{2} \right]$

* Sum of even n no. $\therefore n \times n+1$

* Sum of odd n no. $\therefore (n)^2$

* $\frac{x^n - a^n}{x - a} \Rightarrow n$ is always a natural no.

* $\frac{x^n - a^n}{x + a} \Rightarrow n$ is an even natural no.

* $\frac{x^n + a^n}{x + a} \Rightarrow n$ is an odd natural no.



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Introduction to Number Theory:-

In number theory, the numbers are classified into different types, such as natural numbers, whole numbers, complex numbers, and so on. The sub-classifications of the natural number are given below:

- Odd Numbers – 1, 3, 5, 7, 9, 11, 13, 15, 17, 19.....
- Even Numbers – 2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22 . . .
- Square Numbers – 4, 9, 16, 25, 36, 49, 64, 81, 100 . . .
- Cube Numbers – 8, 27, 64, 125, 216, 343, 512 . . .
- Prime Numbers – 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61 . . .
- Composite Numbers – 4, 6, 8, 9, 10, 12, 14, 15, 16, 18, 20, 21, 22, 24 . . .
- 1 (modulo 4) Numbers – 1, 5, 9, 13, 17, 21, 25, . . .
- 3 (modulo 4) Numbers – 3, 7, 11, 15, 19, 23, 27, . . .
- Triangular Numbers – 3, 6, 10, 15, 21, 28, 36, 45, . . .
- Perfect Numbers – 6, 28, 496, 8128, . . .
- Fibonacci Numbers -1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89. . .

BODMAS:-

BODMAS - Order of Simplification of an expression of numbers

B	=	Bracket
O	=	Of
D	=	Division
M	=	Multiplication
A	=	Addition
S	=	Subtraction



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NUMBER SYSTEM

Q 1. What is the unit digit of the product of $207 \cdot 781 \cdot 39 \cdot 94$?

- (a) 9
- (b) 1
- (c) 7
- (d) 2

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NUMBER SYSTEM

Q 1. What is the unit digit of the product of $207 \cdot 781 \cdot 39 \cdot 94$?

(a) 9

(b) 1

(c) 7

(d) 2

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NUMBER SYSTEM

Q 2. What will come in the place of unit digit in the value of $(7^{35}) \cdot (3^{71}) \cdot (11^{55})$?

- (a) 0
- (b) 3
- (c) 1
- (d) 6

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NUMBER SYSTEM

Q 2. What will come in the place of unit digit in the value of $(7^{35}) \cdot (3^{71}) \cdot (11^{55})$?

(a) 0

(b) 3

(c) 1

(d) 6

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NUMBER SYSTEM

Q 3. Find the number of zeros at the end of the product of $1 \times 2 \times 3 \times 4 \times 5 \times 6$
..... $\times 99 \times 100$:

- (a) 22
- (b) 24
- (c) 26
- (d) 28

NUMBER SYSTEM

Q 3. Find the number of zeros at the end of the product of $1 \times 2 \times 3 \times 4 \times 5 \times 6$
..... $\times 99 \times 100$:

(a) 22

(b) 24

(c) 26

(d) 28

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NUMBER SYSTEM

Q 4. Find the number of zeros at the end of the product of $2 \times 4 \times 6 \times 8 \times 10 \times$
..... $\times 98 \times 100$:

- (a) 10
- (b) 11
- (c) 12
- (d) 15

NUMBER SYSTEM

Q 4. Find the number of zeros at the end of the product of $2 \times 4 \times 6 \times 8 \times 10 \times$
..... $\times 98 \times 100$:

- (a) 10
- (b) 11
- (c) 12**
- (d) 15

NUMBER SYSTEM

Q 5. Find the number of zeros at the end of the product of 10×20
 $\times 30 \times \dots \times 2000$:

- (a) 222
- (b) 249
- (c) 226
- (d) 220

NUMBER SYSTEM

Q 5. Find the number of zeros at the end of the product of 10×20
 $\times 30 \times \dots \times 2000$:

(a) 222

(b) 249

(c) 226

(d) 220

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NUMBER SYSTEM

Q 6. Find the number of factors of 100 :

(a) 8

(b) 9

(c) 10

(d) 12


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NUMBER SYSTEM

Q 6. Find the number of factors of 100 :

(a) 8

(b) 9

(c) 10

(d) 12

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NUMBER SYSTEM

Q 7. Find the number of factors of 80 :

- (a) 10
- (b) 12
- (c) 6
- (d) 8



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NUMBER SYSTEM

Q 7. Find the number of factors of 80 :

- (a) **10**
- (b) 12
- (c) 6
- (d) 8



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NUMBER SYSTEM

Q 8. Find the sum of the factors of 100 :

(a) 127

(b) 217

(c) 219

(d) 189


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NUMBER SYSTEM

Q 8. Find the sum of the factors of 100 :

(a) 127

(b) 217

(c) 219

(d) 189


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NUMBER SYSTEM

Q 9. Find the sum of the factors of 50 :

- (a) 92
- (b) 93
- (c) 97
- (d) 91



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Q 9. Find the sum of the factors of 50 :

(a) 92

(b) 93

(c) 97

(d) 91



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Q 10. Find the average of the factors of 60 :

- (a) 12
- (b) 13
- (c) 14
- (d) 16



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Q 10. Find the average of the factors of 60 :

(a) 12

(b) 13

(c) 14

(d) 16

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Q 11. Find the product of the factors of 100 :

- (a) 10^9
- (b) $10^{9/2}$
- (c) $10^{11/2}$
- (d) 10^{19}

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NUMBER SYSTEM

Q 11. Find the product of the factors of 100 :

- (a) 10^9
- (b) $10^{9/2}$
- (c) $10^{11/2}$
- (d) 10^{19}

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NUMBER SYSTEM

Q 12. How many 3 digit numbers are completely divisible by 6?

- (a) 149
- (b) 150
- (c) 151
- (d) 166


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NUMBER SYSTEM

Q 12. How many 3 digit numbers are completely divisible by 6?

(a) 149

(b) 150

(c) 151

(d) 166

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Q 13. How many 3 digit numbers are completely divisible by 3 and 4 :

- (a) 67
- (b) 75
- (c) 57
- (d) 83



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NUMBER SYSTEM

Q 13. How many 3 digit numbers are completely divisible by 3 and 4 :

(a) 67

(b) 75

(c) 57

(d) 83

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Q 14 . What will be the remainder when 17^{200} is divided by 18?

- (a) 17
- (b) 16
- (c) 1
- (d) 2

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NUMBER SYSTEM

Q 14 . What will be the remainder when 17^{200} is divided by 18?

(a) 17

(b) 16

(c) 1

(d) 2

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NUMBER SYSTEM

Q 14 A. Find the remainder when 3^{85} is divided by 6.

- (a) 3
- (b) 16
- (c) 1
- (d) 2

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NUMBER SYSTEM

Q 14 A. Find the remainder when 3^{85} is divided by 6.

- (a) 3
- (b) 16
- (c) 1
- (d) 2

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Q 14 B. Find the remainder when 2^{70} is divided by 96.

- (a) 64
- (b) 63
- (c) 1
- (d) 2

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Q 14 B. Find the remainder when 2^{70} is divided by 96.

- (a) **64**
- (b) 63
- (c) 1
- (d) 2



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Q 15. What will be the remainder when $(67^{67} + 67)$ is divided by 68?

- (a) 1
- (b) 66
- (c) 67
- (d) 60

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NUMBER SYSTEM

Q 15. What will be the remainder when $(67^{67} + 67)$ is divided by 68?

- (a) 1
- (b) 66**
- (c) 67
- (d) 60

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NUMBER SYSTEM

Q 16. Which of the following number will completely divide $(49^{15} - 1)$?

- (a) 8
- (b) 14
- (c) 51
- (d) 50

NUMBER SYSTEM

Q 16. Which of the following number will completely divide $(49^{15} - 1)$?

- (a) 8
- (b) 14
- (c) 51
- (d) 50

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NUMBER SYSTEM

Q 17. A number when divided by 6 leaves a remainder of 3. When the square of the number is divided by 6, the remainder is:

- (a) 0
- (b) 1
- (c) 3
- (d) 2

NUMBER SYSTEM

Q 17. A number when divided by 6 leaves a remainder of 3. When the square of the number is divided by 6, the remainder is:

- (a) 0
- (b) 1
- (c) 3**
- (d) 2

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NUMBER SYSTEM

Q 18. A number when divided successively by 4 and 5 leaves remainders 1 and 4 respectively. When it is successively divided by 5 and 4, then the respective remainders will be :

- (a) 1, 2
- (b) 2, 3
- (c) 3, 2
- (d) 4, 1

NUMBER SYSTEM

Q 18. A number when divided successively by 4 and 5 leaves remainders 1 and 4 respectively. When it is successively divided by 5 and 4, then the respective remainders will be :

(a) 1, 2

(b) 2, 3

(c) 3, 2

(d) 4, 1

NUMBER SYSTEM

Q 19. A number was divided successively in order by 4, 5, and 6. The remainder were respectively 2, 3, and 4. The number is :

- (a) 214
- (b) 476
- (c) 954
- (d) 1908

NUMBER SYSTEM

Q 19. A number was divided successively in order by 4, 5, and 6. The remainder were respectively 2, 3, and 4. The number is :

- (a) **214**
- (b) 476
- (c) 954
- (d) 1908

NUMBER SYSTEM

Q 20. Which one of the following numbers will completely divide $(4^{61} + 4^{62} + 4^{63} + 4^{64})$?

- (a) 3
- (b) 9
- (c) 11
- (d) 17

NUMBER SYSTEM

Q 20. Which one of the following numbers will completely divide $(4^{61} + 4^{62} + 4^{63} + 4^{64})$?

- (a) 3
- (b) 9
- (c) 11
- (d) 17**

NUMBER SYSTEM

Q 21. Which one of the following numbers will completely divide $5^{51} + 5^{52} + 5^{53}$?

- (a) 11
- (b) 12
- (c) 31
- (d) 32

NUMBER SYSTEM

Q 21. Which one of the following numbers will completely divide $5^{51} + 5^{52} + 5^{53}$

(a) 11

(b) 12

(c) 31

(d) 32

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NUMBER SYSTEM

Q 22. Which one of the following is the common factor of $(47^{43} + 43^{43})$ and $(47^{47} + 43^{47})$?

(a) $47 - 43$

(b) $47 + 43$

(c) $47^{43} + 43^{43}$

(d) $47^{47} + 43^{47}$

NUMBER SYSTEM

Q 22. Which one of the following is the common factor of $(47^{43} + 43^{43})$ and $(47^{47} + 43^{47})$?

(a) $47 - 43$

(b) $47 + 43$

(c) $47^{43} + 43^{43}$

(d) $47^{47} + 43^{47}$

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NUMBER SYSTEM

Q 23. Which one of the following numbers is completely divisible by 99?

- (a) 3572
- (b) 13595
- (c) 913464
- (d) 114345



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NUMBER SYSTEM

Q 23. Which one of the following numbers is completely divisible by 99?

(a) 3572

(b) 13595

(c) 913464

(d) 114345


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NUMBER SYSTEM

Q 24. Which one of the following numbers is completely divisible by 45?

(a) 181560

(b) 331145

(c) 202860

(d) 203350



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NUMBER SYSTEM

Q 24. Which one of the following numbers is completely divisible by 45?

(a) 181560

(b) 331145

(c) 202860

(d) 203350



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NUMBER SYSTEM

Q 25. The sum of digits of a two-digit number is 7. If the digits of the number are interchanged, the number so formed is greater than the original number by 27. Find the original number :

- (a) 29
- (b) 25
- (c) 79
- (d) 32
- (e) None of these

NUMBER SYSTEM

Q 25. The sum of digits of a two-digit number is 7. If the digits of the number are interchanged, the number so formed is greater than the original number by 27. Find the original number :

- (a) 29
- (b) 25**
- (c) 79
- (d) 32
- (e) None of these

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NUMBER SYSTEM $x + y = 7$

Q 12. The sum of digits of a two-digit number is 7. If the digits of the number are interchanged, the number so formed is greater than the original number by 27. Find the original number :

(a) 29
(b) 25
(c) 79
(d) 32
(e) None of these

$x + y = 7$

$10y + x - (10x + y) = 27$

$9y - 9x = 27$

$y - x = 3$

$y + x = 7$

$y - x = 3$

$2y = 10$

$y = 5$

$x = 2$

Original number = 25

NUMBER SYSTEM

Q 26. What is the digit in the blank space of the number $34*7$ so that the number is divisible by 11?

- (a) 3
- (b) 6
- (c) 7
- (d) 8

NUMBER SYSTEM

Q 26. What is the digit in the blank space of the number $34*7$ so that the number is divisible by 11?

- (a) 3
- (b) 6
- (c) 7
- (d) 8**

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NUMBER SYSTEM

Q 27. If the sum of, the digits of a two-digit number and the number formed by reversing its digit is 99, what is the sum of the digits of the original number?

- (a) 9
- (b) 8
- (c) 11
- (d) 10

NUMBER SYSTEM

Q 27. If the sum of, the digits of a two-digit number and the number formed by reversing its digit is 99, what is the sum of the digits of the original number?

- (a) 9
- (b) 8
- (c) 11
- (d) 10

NUMBER SYSTEM

Q 28. If the sum of, the digits of a two-digit number and the number formed by reversing its digits is N , Which one of the following numbers will completely divide N ?

- (a) 9
- (b) 7
- (c) 11
- (d) 18

NUMBER SYSTEM

Q 28. If the sum of, the digits of a two-digit number and the number formed by reversing its digits is N , Which one of the following numbers will completely divide N ?

- (a) 9
- (b) 7
- (c) 11**
- (d) 18

NUMBER SYSTEM

Q 29. If the difference between, a two-digit number and a number formed by reversing its digit is N , Which one of the following numbers will completely divide N :

- (a) 9
- (b) 7
- (c) 11
- (d) 5

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NUMBER SYSTEM

Q 29. If the difference between, a two-digit number and a number formed by reversing its digit is N , Which one of the following numbers will completely divide N :

- (a) 9
- (b) 7
- (c) 11
- (d) 5

NUMBER SYSTEM

Q 30. If the difference between a two digit number and the number formed by reversing its digit is 45, what is the difference between the digits of the original number :

- (a) 9
- (b) 2
- (c) 5
- (d) 1

NUMBER SYSTEM

Q 30. If the difference between a two digit number and the number formed by reversing its digit is 45, what is the difference between the digits of the original number :

- (a) 9
- (b) 2
- (c) 5**
- (d) 1

NUMBER SYSTEM

Q 31. A 4-digit number is formed by repeating a 2-digit number such as 2525, 3232, etc. Any number of this form is always divisible by :

- (a) Smallest two-digit prime number
- (b) 7
- (c) 13
- (d) smallest three-digit prime number

NUMBER SYSTEM

Q 31. A 4-digit number is formed by repeating a 2-digit number such as 2525, 3232, etc. Any number of this form is always divisible by :

(a) Smallest two-digit prime number

(b) 7

(c) 13

(d) smallest three-digit prime number

NUMBER SYSTEM

Q 32. $7^{12} - 4^{12}$ is exactly divisibly by which of the following?

- (a) 36
- (b) 35
- (c) 34
- (d) 33



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NUMBER SYSTEM

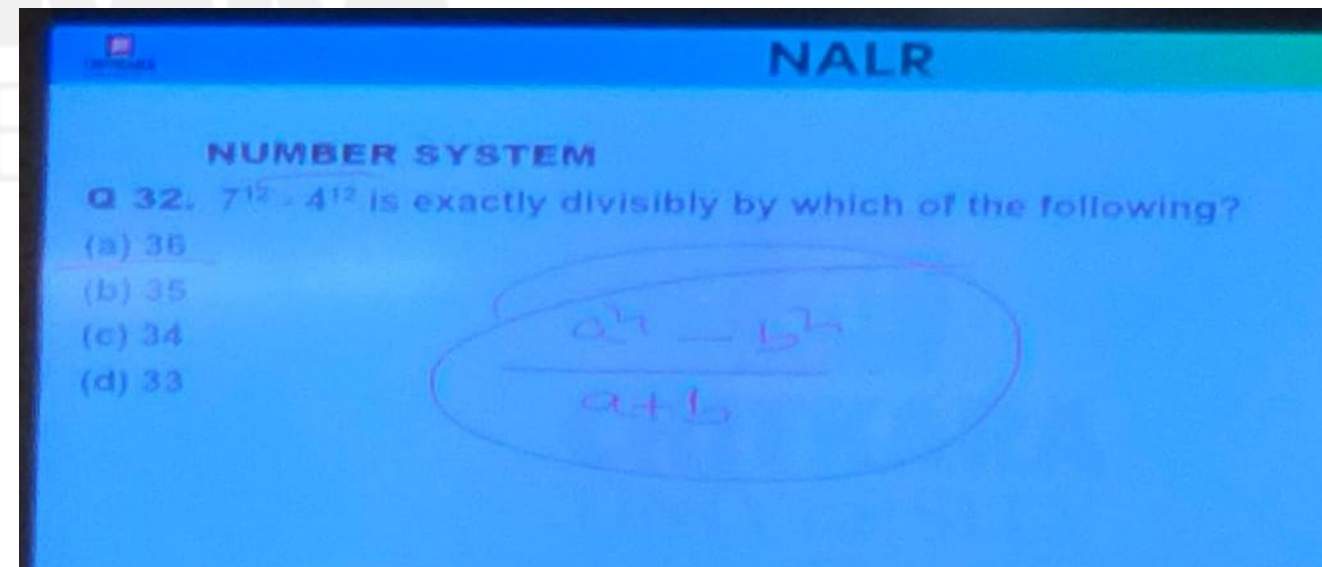
Q 32. $7^{12} - 4^{12}$ is exactly divisibly by which of the following?

(a) 36

(b) 35

(c) 34

(d) 33



NUMBER SYSTEM

Q 33. Find the sum of the first fifty natural numbers :

- (a) 1144
- (b) 1275
- (c) 1325
- (d) 1075



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NUMBER SYSTEM

Q 33. Find the sum of the first fifty natural numbers :

(a) 1144

(b) 1275

(c) 1325

(d) 1075

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NUMBER SYSTEM

Q 34. Find the value of $51+52+53+54+\dots\dots\dots+100$:

- (a) 2443
- (b) 1754
- (c) 2673
- (d) 3775

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NUMBER SYSTEM

Q 34. Find the value of $51+52+53+54+\dots\dots\dots+100$:

(a) 2443

(b) 1754

(c) 2673

(d) 3775

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NUMBER SYSTEM

Q 35. Find the sum of the squares of the first 30 natural numbers :

- (a) 9455
- (b) 8372
- (c) 7849
- (d) 6973

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NUMBER SYSTEM

Q 35. Find the sum of the squares of the first 30 natural numbers :

- (a) **9455**
- (b) 8372
- (c) 7849
- (d) 6973

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NUMBER SYSTEM

Q 36. Find the value of $2^2 + 4^2 + 6^2 + 8^2 + \dots + 20^2$:

- (a) 2870
- (b) 1321
- (c) 1540
- (d) 1550

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NUMBER SYSTEM

Q 36. Find the value of $2^2 + 4^2 + 6^2 + 8^2 + \dots + 20^2$:

- (a) 2870
- (b) 1321
- (c) 1540**
- (d) 1550

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NUMBER SYSTEM

Q37. Find the value of $1^2 + 3^2 + 5^2 + 7^2 + \dots + 19^2$:

- (a) 1335
- (b) 1330
- (c) 1332
- (d) 1334

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NUMBER SYSTEM

Q37. Find the value of $1^2 + 3^2 + 5^2 + 7^2 + \dots + 19^2$:

(a) 1335

(b) 1330

(c) 1332

(d) 1334

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NUMBER SYSTEM

Q38. If $1^2 + 2^2 + 3^2 + 4^2 + \dots + 10^2 = 385$, find the value of $2^2 + 4^2 + 6^2 + \dots + 20^2$:

- (a) 1250
- (b) 1540
- (c) 1190
- (d) 1375

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NUMBER SYSTEM

Q38. If $1^2 + 2^2 + 3^2 + 4^2 + \dots + 10^2 = 385$, find the value of $2^2 + 4^2 + 6^2 + \dots + 20^2$:

(a) 1250

(b) 1540

(c) 1190

(d) 1375

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NUMBER SYSTEM

Q 39. Find the value of $11^2 + 12^2 + 13^2 + 14^2 + \dots + 20^2$:

- (a) 2870
- (b) 2485
- (c) 2670
- (d) 2495

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NUMBER SYSTEM

Q 39. Find the value of $11^2 + 12^2 + 13^2 + 14^2 + \dots + 20^2$:

(a) 2870

(b) 2485

(c) 2670

(d) 2495

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NUMBER SYSTEM

Q 40. Find the value of $1^3 + 3^3 + 5^3 + 7^3 + \dots + 29^3$:

- (a) 36100
- (b) 101025
- (c) 32500
- (d) 44700

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NUMBER SYSTEM

Q 40. Find the value of $1^3 + 3^3 + 5^3 + 7^3 + \dots + 29^3$:

(a) 36100

(b) 101025

(c) 32500

(d) 44700

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NUMBER SYSTEM

Q 24. Find the value of $1^3 + 3^3 + 5^3 + 7^3 + \dots + 29^3$:

(a) 36100
(b) 101025
(c) 32500
(d) 44700

$$(n+1)^2 \left[(n+1)^2 - 1 \right]$$

$$8$$

NUMBER SYSTEM

Q 41. If $1^3 + 2^3 + 3^3 + 4^3 + \dots + 10^3 = 3025$, find the value of $2^3 + 4^3 + 6^3 + \dots + 20^3$:

- (a) 2875
- (b) 24200
- (c) 3080
- (d) 39400

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Q 25. If $1^3 + 2^3 + 3^3 + 4^3 + \dots + 10^3 = 3025$, find the value of $2^3 + 4^3 + 6^3 + \dots + 20^3$:

- (a) 2875
- (b) 2420
- (c) 3080
- (d) 39400

$$2 \times \left(\frac{n}{2}\right) \left[\left(\frac{n}{2} + 1\right)^2\right]$$

NUMBER SYSTEM

Q 41. If $1^3 + 2^3 + 3^3 + 4^3 + \dots + 10^3 = 3025$, find the value of $2^3 + 4^3 + 6^3 + \dots + 20^3$:

(a) 2875

(b) 24200

(c) 3080

(d) 39400

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NUMBER SYSTEM

Q 42. Find the sum of all even numbers up to 100 :

- (a) 2295
- (b) 2425
- (c) 2495
- (d) 2550



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Q 42. Find the sum of all even numbers up to 100 :

(a) 2295

(b) 2425

(c) 2495

(d) 2550

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Q 43. Find the sum of the all odd number up to 100 :

- (a) 2100
- (b) 2500
- (c) 2300
- (d) 2200



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Q 43. Find the sum of the all odd number up to 100 :

(a) 2100

(b) 2500

(c) 2300

(d) 2200


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Q 44. Find the number of prime factors of $6^{20} \cdot 11^{11} \cdot 21^{21}$:

- (a) 83
- (b) 93
- (c) 103
- (d) 113

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NUMBER SYSTEM

Q 44. Find the number of prime factors of $6^{20} \cdot 11^{11} \cdot 21^{21}$:

- (a) 83
- (b) 93**
- (c) 103
- (d) 113

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NUMBER SYSTEM

Q 45. Find the number of prime factors of $14^{14} \cdot 15^{15}$:

- (a) 48
- (b) 58
- (c) 68
- (d) 78

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NUMBER SYSTEM

Q 45. Find the number of prime factors of $14^{14} \cdot 15^{15}$:

(a) 48

(b) 58

(c) 68

(d) 78

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NUMBER SYSTEM

Q 46. What will be the remainder when $(27^{27} + 17^{27})$ is divided by 11?

- (a) 27
- (b) 17
- (c) 0
- (d) 1

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NUMBER SYSTEM

Q 46. What will be the remainder when $(27^{27} + 17^{27})$ is divided by 11?

(a) 27

(b) 17

(c) 0

(d) 1

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Q 47. If n is a natural number, $(n^3 - n)$ will always be divisible by:

- (a) 6 only
- (b) 6 and 12 both
- (c) 12 only
- (d) by 18 only

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NUMBER SYSTEM

Q 47. If n is a natural number, $(n^3 - n)$ will always be divisible by:

- (a) **6 only**
- (b) 6 and 12 both
- (c) 12 only
- (d) by 18 only

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NUMBER SYSTEM

Q 48. ($x^n - a^n$) is completely divisible by $(x - a)$, when

- (a) n is any natural number
- (b) n is an even natural number
- (c) n is an odd natural number
- (d) n is a prime number

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NUMBER SYSTEM

Q 48. ($x^n - a^n$) is completely divisible by $(x - a)$, when

(a) n is any natural number

(b) n is an even natural number

(c) n is an odd natural number

(d) n is a prime number

The screenshot shows a presentation slide with a blue header containing the logos of Chitkara University and the Quantitative Aptitude Center, and the text 'NALR'. The main content area is white and contains the title 'NUMBER SYSTEM' and the question 'Q 29. ($x^n - a^n$) is completely divisible by $(x - a)$, when'. Below the question are four options: (a) n is any natural number, (b) n is an even natural number, (c) n is an odd natural number, and (d) n is a prime number. Option (a) is underlined in red. To the right of the options, there is a handwritten note in red ink that says 'n is any natural number' with an arrow pointing to option (a). The slide also features a large, faint watermark of the Chitkara University logo in the background.

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NALR

NUMBER SYSTEM

Q 29. ($x^n - a^n$) is completely divisible by $(x - a)$, when

(a) n is any natural number

(b) n is an even natural number

(c) n is an odd natural number

(d) n is a prime number

$x^n - a^n$ — natural
 $x - a$ —

NUMBER SYSTEM

Q 49. ($x^n - a^n$) is completely divisible by $(x + a)$, when

- (a) n is any natural number
- (b) n is an even natural number
- (c) n is an odd natural number
- (d) n is a prime number

CHITKARA
UNIVERSITY

NUMBER SYSTEM

Q 49. ($x^n - a^n$) is completely divisible by $(x + a)$, when

(a) n is any natural number

(b) n is an even natural number

(c) n is an odd natural number

(d) n is a prime number

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NUMBER SYSTEM

Q 50. $(x^n + a^n)$ is completely divisible by $(x + a)$, when

- (a) n is any natural number
- (b) n is an even natural number
- (c) n is an odd natural number
- (d) n is a prime number

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NUMBER SYSTEM

Q 50. $(x^n + a^n)$ is completely divisible by $(x + a)$, when

- (a) n is any natural number
- (b) n is an even natural number
- (c) n is an odd natural number**
- (d) n is a prime number

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NUMBER SYSTEM

Q 51. Which one of the following is a prime number?

- (a) 161
- (b) 221
- (c) 373
- (d) 437

NUMBER SYSTEM

Q 51. Which one of the following is a prime number?

- (a) 161
- (b) 221
- (c) 373**
- (d) 437



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NUMBER SYSTEM

Q 52. Which one of the following is a prime number?

- (a) 119
- (b) 187
- (c) 247
- (d) 71

NUMBER SYSTEM

Q 52. Which one of the following is a prime number?

(a) 119

(b) 187

(c) 247

(d) 71



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NUMBER SYSTEM

Q 53. Find the largest four-digit number which is divisible by 88 :

- (a) 9944
- (b) 9768
- (c) 9988
- (d) 8888

CHITKARA
UNIVERSITY

NUMBER SYSTEM

Q 53. Find the largest four-digit number which is divisible by 88 :

- (a) **9944**
- (b) 9768
- (c) 9988
- (d) 8888

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NUMBER SYSTEM

Q 54. If a number is divided by 111, the remainder is 31. What will be the remainder if it is divided by 37?

- (a) 31
- (b) 32
- (c) 33
- (d) 0

NUMBER SYSTEM

Q 54. If a number is divided by 111, the remainder is 31. What will be the remainder if it is divided by 37?

- (a) 31
- (b) 32
- (c) 33
- (d) 0

NUMBER SYSTEM

Q 55. On multiplying a number by 7, the product is a number made of only the digit 3. The smallest such number is:

- (a) 47619
- (b) 47719
- (c) 48619
- (d) 47649

NUMBER SYSTEM

Q 55. On multiplying a number by 7, the product is a number made of only the digit 3. The smallest such number is:

- (a) **47619**
- (b) 47719
- (c) 48619
- (d) 47649



THANK YOU

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